Unsupervised 3D Masked-Autoencoder for Coastal Ocean Anomaly Detection

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Abstract

Signs of harmful algal blooms along the U.S. Pacific Northwest coast are visible in satellite imagery, but predicting downstream biology from single channels like chlorophyll-a has worked poorly. We train a 3D convolutional masked-autoencoder on 22 years (2003–2024) of MODIS Aqua imagery over the 41–49° N coastal box. It takes four ocean channels over a 4-day window and produces one unsupervised "anomaly score" per region per day from its reconstruction error. We evaluate the score on three independent checks: a 7-day horizon where it was the only method we tested with positive R^2 in every region (0.10–0.26) while a matched-complexity PCA baseline stays at or below zero in every region. The score for the region containing the NEMO mooring tracks in-situ Pseudo-nitzschia cell density measured there during 2016–2018 (Spearman ρ=+0.28, p=0.02; Pearson r=+0.46) though the particulate-toxin signal is weaker. Its summer-mean score for 2014 is the highest on record in three of four regions, matching the documented Pacific marine heatwave.

1. Introduction

Toxin-producing *Pseudo-nitzschia* blooms happen every year along the U.S. Pacific Northwest (PNW) coast. The toxin they make, domoic acid, builds up in razor clams and shuts down beach harvests [10, 17]. These blooms are not completely invisible from space, since satellites can measure the elevated chlorophyll and so forth. The problem is that picking out which patches of water will actually produce toxin from those satellite channels has historically been hard [1, 5], with only some blooms actually producing toxin.

In addition, single MODIS pixels near the beach are noisy as clouds, sun glint, river plumes, and shallow-water effects all can corrupt the signal. The intuition behind this paper is that a per-region aggregate should in theory be cleaner than any single pixel, and that a model can learn which spatial structures to aggregate over. Masked autoencoders [4] are a self-supervised approach where the model has to fill in chunks of the image that have been hidden, and in natural images this forces the model to learn higher-level structure rather than just local texture. We adapt that idea to coastal satellite imagery.

Our main question revolves around whether an unsupervised representation of the satellite imagery, trained with no bloom labels, yields a per-region anomaly score that (i) forecasts next-week ocean state better than matched-complexity baselines, (ii) agrees with in-situ bloom biology at a coastal mooring, and (iii) recovers documented PNW climate events.

2. Related Work

Operational monitoring has used satellite chlorophyll-a as a near-real-time bloom indicator for coastal HABs [11], but in PNW coastal water the relationship between chl-a and beach-level toxin outcomes is much weaker [16, 17]. Multi-channel approaches that combine SST, fluorescence, and clarity perform incrementally better [1], but any per-pixel single-day framing will remain limiting as blooms are a region wide phenomena.

In Earth observation, SatMAE [3] applies the masked-autoencoder principle [4] to multi-spectral satellite imagery for land-cover classification, and self-supervised pretraining is now a broad subfield of remote sensing [18]. We adapt it to a 3D (time-augmented) coastal ocean cube, where the temporal axis lets the model learn dynamics as well as spatial structure important in any amount of future forecasting.

The Environmental Sample Processor [15] is an autonomous in-water sampler deployed at the NEMO-Ćhá?ba mooring [13], measuring *PN* cell density and particulate domoic acid daily. Moore et al. [13] find that satellite chlorophyll-a at NEMO tracks bulk *PN* biomass but fails to serve as a proxy for the actual toxin levels that matter, which is part of why we use a multi-channel learned representation rather than a single-channel retrieval.

3. Method

Data. The training cube is built from MODIS Aqua Level-3 daily ocean-color products [14] covering 2003–2024 (∼4,700 daily rolling 8-day composites) on a 321×409

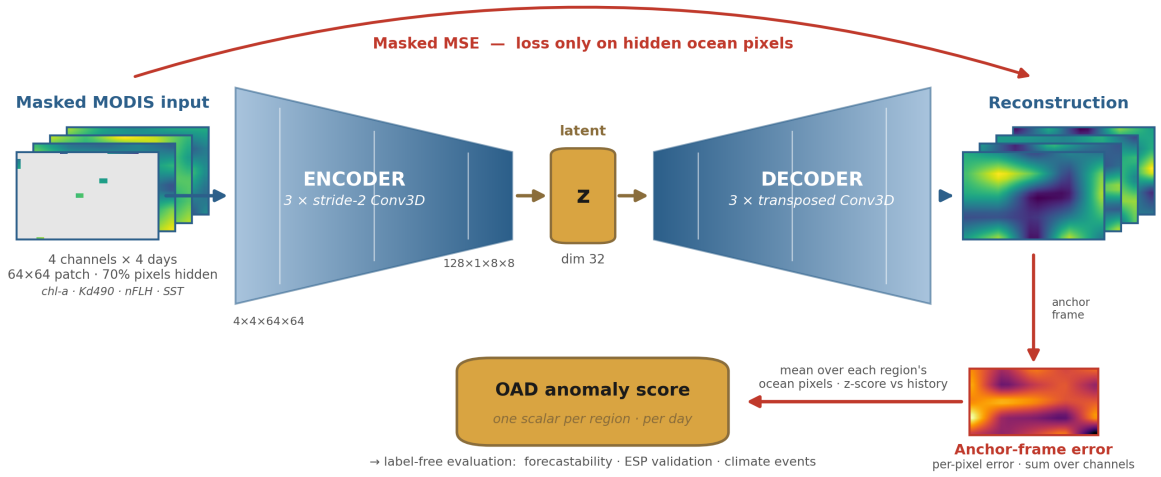


Figure 1. OAD pipeline: a 3D masked-autoencoder reconstructs a 4-channel, 4-day MODIS patch, and the per-region reconstruction error becomes the daily anomaly score.

grid at 0.025° resolution over the 41–49° N PNW coastal box (Fig. 2). Four channels are used: chl-a, Kd490 (water clarity), normalized fluorescence (nFLH), and SST. A 2D mask excludes land pixels, and NaN pixels are excluded per-pixel from the masked-MSE reconstruction loss.

Architecture. OAD is a 3D ConvAE applied to 64 × 64 spatial patches tiled over the 321 × 409 grid, each of shape $(C{=}4, T{=}4, H{=}64, W{=}64)$ (Fig. 1). The encoder applies three 3D-convolution blocks (channel widths $4{\to}32{\to}64{\to}128$) with stride-2 spatial downsampling, batch normalization, and ReLU activation, collapsing each patch to a $128{\times}1{\times}8{\times}8$ feature map; a fully-connected layer then compresses this to a 32-dimensional latent code. The decoder inverts these steps: a fully-connected layer expands the latent back to the $128{\times}1{\times}8{\times}8$ map, followed by three transposed-convolution blocks that upsample to the original patch.

Training. We train the network as a masked autoencoder [4]: a random fraction of the input pixels is hidden and the network must reconstruct them from the surrounding context, which keeps it from simply copying local texture. A plain autoencoder without masking did exactly that, and its reconstruction error tracked bloom events poorly. We hide 70% of pixels (`mae070`) in the headline model, a high masking ratio that yields positive multi-step forecast skill in all five regions at the 7-day lead (Sec. 4.1).

For each daily composite at region R, the trained model produces a per-pixel reconstruction. We average per-pixel reconstruction error over in-region ocean pixels and z-score the result against an in-region historical distribution, producing a single scalar per (region, day). Downstream analyses use only this per-region daily scalar; the convolutional model is not run again afterward.

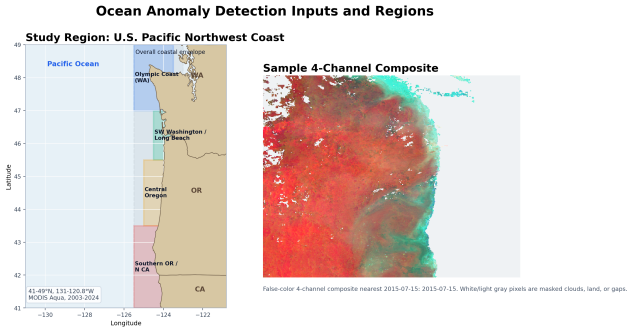


Figure 2. The U.S. PNW study region (41–49° N): the overall coastal envelope plus four alongshore sub-regions (Olympic Coast, SW Washington/Long Beach, Central Oregon, Southern OR/N. CA), shown over an example MODIS Aqua 8-day false-color composite. Light-gray pixels are cloud, land, or gaps.

4. Results

The OAD score is evaluated in three independent ways: lead-time forecastability against baselines (Sec. 4.1), in-situ biological validation at the NEMO mooring (Sec. 4.2), and consistency with documented PNW climate events (Sec. 4.3).

4.1. Lead-time forecastability vs. baselines

We compare OAD against two baselines, both forecasting per-region anomaly state from the past four days on the 2019+ held-out period. The PCA baseline projects the same four-channel regional input into the same number of components as the autoencoder bottleneck, then fits a linear regression from those components to the target score at each lead time; it is a matched-complexity linear control. The climatology baseline predicts next week's score as the historical mean for that day-of-year, ignoring all recent observations. Tab. 1 reports results for SW Washington as a representative region, and Fig. 3 shows the cross-region picture at the 7-day horizon.

Table 1. Lead-time forecastability R^2, SW Washington region (2019+ held-out); the 1-day value is inflated by 8-day composite overlap.

Lead	OAD-50	OAD-70	PCA	Climatology
1 day	+0.84	+0.87	−0.13	+0.71
7 days	+0.08	+0.15	−0.13	−0.10
14 days	+0.01	+0.05	−0.13	−0.19

At the honest 7-day lead, OAD is the only method we tested with positive R^2 in every one of the five PNW regions (range 0.10–0.26). PCA, with the same number of dimensions, stays at or below zero in every region, and the day-of-year climatology baseline actively does worse than predicting the mean in most regions (Fig. 3). These values are low, but only OAD clears zero, while the matched-complexity baselines do not. A further diagnostic is visible in Tab. 1: unlike OAD and climatology, whose 1-day R^2 values are elevated by the 8-day rolling composite overlap, PCA's score is equally poor at every lead time. This flatness reveals that PCA's anomaly score carries no temporal autocorrelation for the forecasting model to exploit at any horizon; the advantage OAD shows at 7 days therefore reflects learned temporal structure, not input-level smoothing.

The right test of whether the learning matters, rather than just the regional aggregation, is whether OAD outperforms a matched-complexity linear baseline. PCA is that baseline: it receives the same four-channel input and uses the same bottleneck dimension as the autoencoder, so any gap over PCA at the 7-day lead (Fig. 3) must come from what masked-autoencoder pretraining learns, not from having more free parameters. PCA drops to zero or below in every region while OAD is positive everywhere. The autoencoder is supplying the representational step that linear projection cannot.

4.2. In-situ validation against ESP/ChaBa

To test whether OAD captures real biology, we compare it against independent in-situ measurements at the same place and time. We use ESP-derived *PN* cell density and particulate domoic acid (pDA) from the 2016–2018 ChaBa-NEMO mooring (47.97° N, 124.97° W) [13]. NEMO lies within our Olympic Coast region, so we compare them against that one region's OAD score. Because cell-density and toxin distributions are heavy-tailed (a few bloom days dominate), we report both Pearson r on raw values and the rank-based Spearman ρ (Tab. 2), each with a bootstrap confidence interval.

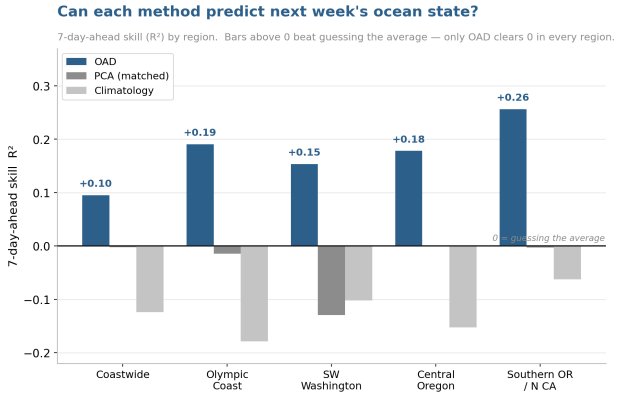


Figure 3. Held-out 7-day-ahead forecast skill (R^2) by region (2019+): OAD (blue) is positive in all five regions (0.10–0.26), while the matched-complexity PCA and day-of-year climatology baselines are at or below zero everywhere.

Table 2. OAD score vs. in-situ ESP measurements at NEMO (2016–2018), with Pearson and rank-based Spearman correlations and 95% bootstrap CIs.

Measurement	Pearson r	Spearman ρ	N
PN cells	**+0.46** [.17,.66]	**+0.28** [.01,.51]	76
pDA (toxin)	+0.33 [−.01,.53]	+0.16 [−.07,.36]	87

The *PN*-cell signal is the robust one. The OAD score for the Olympic Coast region tracks measured cell density with a Pearson CI well above zero and a rank-based bootstrap CI that also excludes zero (Tab. 2). As Fig. 4 shows, median cell density rises monotonically across terciles of the OAD score. The particulate-toxin signal is weaker. The raw Pearson $r = 0.33$ is carried by a handful of high-toxin days, and the rank correlation is not significant (Spearman $\rho = 0.16$, $p = 0.13$, CI spanning zero). We therefore treat *PN* cell density as the validated target and pDA as a suggestive but unconfirmed secondary. The correlation is moderate and the satellite misses plenty of day-to-day biology, but because the model never trained on these in-water measurements, even this agreement is informative.

The time-series overlay (Fig. 5) shows OAD and ESP peaks tending to move together across the five 2016–2018 deployment windows. The 2017 spring deployment is the clearest case, where OAD ramps up several days before the

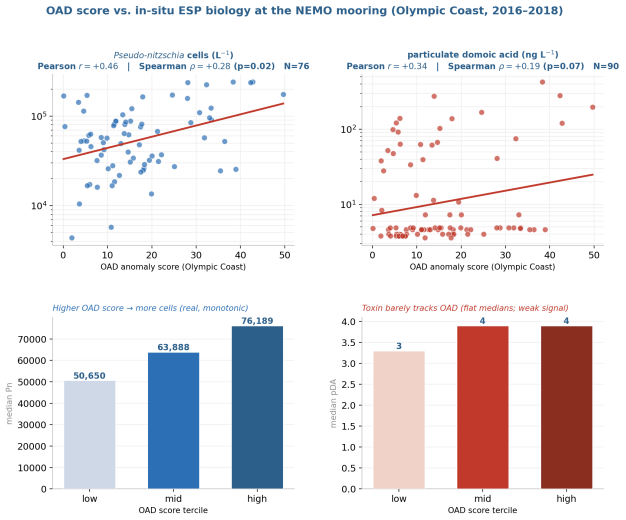


Figure 4. OAD score vs. in-situ biology at NEMO (2016–2018): per-day scatter (top) and median ESP value within OAD-score terciles (bottom), for cell density (left) and toxin (right).

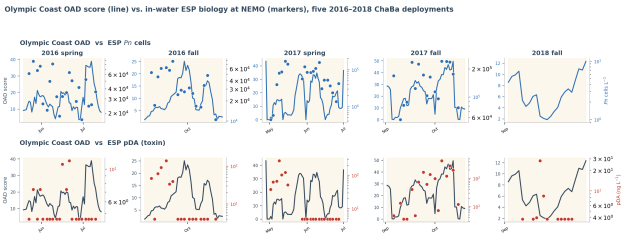


Figure 5. Daily overlay of the Olympic Coast OAD score (line) and ESP measurements (markers): *PN* cells (top) and particulate domoic acid (bottom), across the five 2016–2018 ChaBa deployments (shaded).

ESP-measured *PN* peak, which suggests it may be useful as a leading indicator for when to deploy in-water samplers.

4.3. Agreement with documented PNW climatology

The unsupervised score also recovers well-established PNW coastal ocean patterns. Three stand out.

The score rises sharply in late spring (April–May), peaks in mid-summer, and declines through autumn across all five regions, matching canonical PNW *PN* bloom phenology [12, 16, 17]. Amplitude is largest in SW Washington (Columbia River plume plus shelf upwelling) and Central Oregon (strong seasonal coastal upwelling), with the Juan de Fuca eddy sustaining nutrient-rich water off the Olympic Coast [8].

One test of the score's climate sensitivity is whether it picks out the year of the 2014–16 Pacific marine heatwave ("the Blob" [2]), which triggered the largest *P. australis* bloom on record [9, 10]. Fig. 6 plots the monthly-mean OAD score per region across 2003–2024. In Olympic Coast, Central Oregon, and Southern OR/N. California, the 2014 summer mean (May–September) is the highest across the full 2003–2024 record (excluding the 2009–2011 MODIS gap), and in SW Washington it is the second-highest. The model was never told which years had blooms, so this ranking comes entirely from the satellite imagery itself.

Regional scores move broadly together on monthly timescales, consistent with large-scale Pacific climate drivers (PDO, ENSO) [6, 7], but each region keeps its own short-term signal from local eddies, plumes, and wind events.

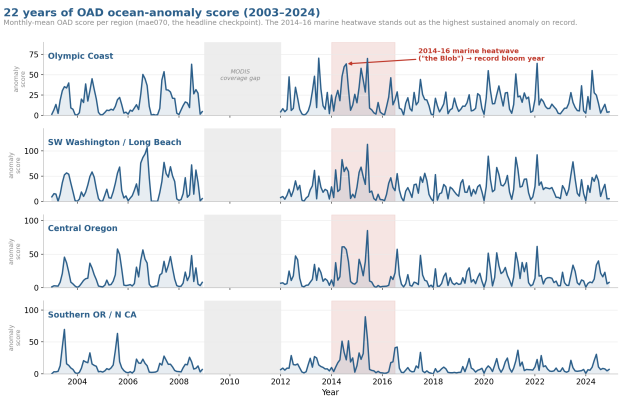


Figure 6. Monthly-mean OAD score per region (2003–2024); the shaded 2014–16 marine heatwave is the highest sustained anomaly and the gray band marks the 2009–11 MODIS coverage gap.

5. Discussion

Although across all three checks the absolute numbers are small, the comparisons still hold. OAD is the only method that beats the mean at a 7-day lead; its Pseudo-Nitzschia cell-density correlation stays significant under a rank test; and it flags 2014 as the standout heatwave year even with no temporal labels. Although none of this proves the score is a finished bloom-prediction tool, it does show that an unsupervised representation of the satellite cube picks up structure the linear baselines miss and can line up with real biology at the source.

For our headline mae070 checkpoint, per-region Pearson correlation between the OAD score and the in-region valid-pixel fraction is moderate ($r \approx 0.37$–0.49 across the five regions, accounting for roughly 14–24% of score variance). The mae050 variant is somewhat lower on this metric, the cloud half of the tradeoff discussed in Sec. 3. Downstream applications should carry the in-region valid-pixel fraction as a parallel feature so consuming models can learn to discount cloudy weeks.

What this paper does not show is whether the source-region OAD score predicts beach-level shellfish toxicity. That depends on a chain of additional physics, including

wind direction, onshore transport, which *PN* species is present and whether it is producing toxin, and how much each clam happens to filter. We test only the offshore signal here.

Limitations. (1) the 2009–2011 MODIS coverage gap leaves wide uncertainty in some regions, (2) the cloud-confound is partial (∼14–24% variance), (3) in-situ validation uses only one mooring (NEMO) with 76–90 daily samples, and (4) regions are hand-defined, so learning regions via clustering on the latent is a natural extension.

Lessons learned. The matched-complexity PCA baseline ended up mattering more than the autoencoder itself: without it, a positive R^2 reads as nothing more than "a bigger model fits better." It was the gap between the two, not OAD's absolute number, that turned out to be the result worth reporting, especially since the correlations came in lower than we expected. The evaluation horizon mattered as much as the metric: the inflated 1-day score from 8-day composite overlap would have overstated everything had we stopped there instead of pushing to a leakage-safe 7-day lead. And the cloud-cover confound only surfaced because we checked for it directly, which is why a learned signal should be tested against the obvious nuisance variable before it is trusted.

6. Conclusion

We trained an unsupervised 3D masked-autoencoder on 22 years of MODIS Aqua imagery and turned its reconstruction error into one anomaly score per coastal region per day. The score is modest but carries real signal: it is the only method we tested that beats the mean at a 7-day lead in every PNW region, it tracks in-situ *PN* cell density at a mooring it never trained on, and it independently picks out the 2014 marine heatwave. Useful next steps are to learn the regions by clustering the latent instead of hand-defining them, to regress out the cloud-cover correlation, to validate against more than the single NEMO mooring, and to test whether the offshore score, combined with onshore-transport physics, adds skill to beach-level toxicity forecasts, the question this paper deliberately leaves open.

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